

Exact chordal separation of spherical Fibonacci points

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Abstract

We prove the exact finite separation conjecture stated in Brauchart's 2012 Oberwolfach contribution: the rational, unshifted spherical Fibonacci configuration with F_n points has minimum Euclidean chord $2/\sqrt{F_n}$ for every $n \geq 3$. A latitude–longitude inequality reduces the lower bound to the classical product separation of Fibonacci lattices. We include an elementary proof of the weaker arithmetic estimate needed here, making the argument self-contained.

1 The conjecture and its precise scope

Let $F_0 = 0$, $F_1 = F_2 = 1$, and $F_{n+1} = F_n + F_{n-1}$. For an integer $n \geq 3$, put $q = F_n$ and $p = F_{n-1}$, and define

$$z_k = \left(2\sqrt{\frac{k}{q} \left(1 - \frac{k}{q}\right)} \cos \frac{2\pi pk}{q}, \quad 2\sqrt{\frac{k}{q} \left(1 - \frac{k}{q}\right)} \sin \frac{2\pi pk}{q}, \quad 1 - \frac{2k}{q} \right), \quad 0 \leq k < q. \quad (1)$$

These are distinct points on the unit sphere. Brauchart's contribution [1, pp. 2436–2439, especially p. 2438] conjectures the following exact minimum, already attained by the north pole and its first latitude point.

Theorem 1. *For every integer $n \geq 3$, the configuration (1) satisfies*

$$\min_{0 \leq i < j < q} \|z_i - z_j\| = \frac{2}{\sqrt{q}}.$$

The unique minimizing unordered pair is $\{z_0, z_1\}$.

Here the metric is the ambient Euclidean chord. Its corresponding minimum geodesic distance is $2 \arcsin(1/\sqrt{q})$. The singleton $n = 2$ has no pair and is excluded. The theorem concerns the displayed rational longitudes and unshifted heights; it makes no assertion about midpoint heights, irrational golden angles, arbitrary cardinalities, discrepancy, or optimal packing among all spherical configurations.

2 The arithmetic and geometric inequalities

The arithmetic ingredient is classical. For the dual Fibonacci lattice $\{(h_1, h_2) \in \mathbb{Z}^2 : h_1 + ph_2 \equiv 0 \pmod{q}\}$, its hyperbolic-cross index, the minimum of $\max(1, |h_1|) \max(1, |h_2|)$ over nonzero vectors, is F_{n-2} ; see Zaremba [3, p. 186] and the explicit modern statement in Suryanarayana, Cools and Nuyens [4, Theorem 2.2]. Applying this to $(-r, m)$ gives $m|r| \geq F_{n-2} \geq q/3$. For completeness, the following proof establishes the needed weaker inequality without relying on that index theorem.

Lemma 2. *If $n \geq 3$, $1 \leq m < q = F_n$, and $\ell \in \mathbb{Z}$, then*

$$m|mp - \ell q| \geq q/3, \quad p = F_{n-1}.$$

Proof. Choose the unique $j \geq 2$ with $F_j \leq m < F_{j+1}$. Then $j \leq n-1$; write $h = n-j \geq 1$. The integer vectors (F_j, F_{j-1}) and (F_{j+1}, F_j) form a unimodular basis, since their determinant is $(-1)^{j-1}$. Consequently

$$(m, \ell) = S(F_j, F_{j-1}) + T(F_{j+1}, F_j), \quad S, T \in \mathbb{Z}.$$

Here $S \neq 0$: otherwise a positive $m < F_{j+1}$ would be a multiple of F_{j+1} . Both S, T positive would give $m \geq F_j + F_{j+1}$; both negative, or $S < 0, T = 0$, would give $m < 0$. Thus either $T = 0, S > 0$ or S, T have opposite signs. The recurrence gives

$$pF_j - qF_{j-1} = (-1)^{j-1}F_h, \quad pF_{j+1} - qF_j = (-1)^jF_{h-1}.$$

It follows that $|mp - \ell q| = |SF_h - TF_{h-1}| \geq F_h$, including the terminal case $h = 1$, where $F_{h-1} = 0$. Finally the Fibonacci addition identity gives

$$q = F_jF_{h+1} + F_{j-1}F_h \leq 3F_jF_h \leq 3m|mp - \ell q|,$$

using $F_{h+1} \leq 2F_h$ and $F_{j-1} \leq F_j$. The determinant, error and addition identities all follow by induction from the defining recurrence. $\square$

Lemma 3. *For two points with latitude parameters $0 \leq a < b \leq 1$ in (1), let $d = b - a$ and let c be the cosine of their longitude difference. Their squared chord D^2 satisfies*

$$c \leq 0 \implies D^2 \geq 4d, \quad 0 < c < 1 \implies D^2 \geq 4d\sqrt{1-c^2}.$$

Proof. Put $U = 2(a + b - 2ab)$ and $V = 4\sqrt{a(1-a)b(1-b)}$. Direct expansion gives

$$D^2 = 2(U - cV), \quad U \geq 2d, \quad V^2 = U^2 - 4d^2, \quad U - 2d = 4a(1-b).$$

For $c \leq 0$ the first bound follows. For $0 < c < 1$, $U - cV > 0$, and the identity

$$(U - cV)^2 - 4d^2(1 - c^2) = (V - cU)^2 \geq 0$$

proves the second bound. No planar distance-preservation property of the equal-area spherical map is used. $\square$

3 Proof of the exact separation

Proof of Theorem 1. Fix $0 \leq i < j < q$ and set $m = j - i$ and $d = m/q$. Choose $\ell \in \mathbb{Z}$ so that $r = mp - \ell q$ satisfies $0 < |r| \leq q/2$. Such a nonzero residue exists because consecutive Fibonacci numbers are coprime and $0 < m < q$. The longitude cosine is $c = \cos(2\pi r/q)$. If $c \leq 0$, Lemma 3 gives $\|z_i - z_j\|^2 \geq 4m/q \geq 4/q$. If $c > 0$, then $0 < |r| < q/4$. Concavity of sine on $[0, \pi/2]$ gives $\sin(2\pi|r|/q) \geq 4|r|/q$, so the two lemmas give

$$\|z_i - z_j\|^2 \geq \frac{4m}{q} \sin \frac{2\pi|r|}{q} \geq \frac{16m|r|}{q^2} \geq \frac{16}{3q} > \frac{4}{q}.$$

This proof includes $q = 2, 3, 5$ without separate numerical cases. Since $z_0 = (0, 0, 1)$, direct calculation gives $\|z_0 - z_1\|^2 = 4/q$, proving the exact minimum. For equality in the nonpositive-cosine case we must have $m = 1$ and $U = 2d$. As $b = j/q < 1$, the identity $U - 2d = 4a(1-b)$ forces $a = 0$. Thus equality occurs only for $i = 0, j = 1$. $\square$

4 Source conventions, priority and verification

The displayed range $0 \leq k < q$ and explicit pole witness in [1, p. 2438] specify the convention used here, despite the following set notation’s endpoint shift. The range $1 \leq k \leq q$ gives an isometric set via $k \mapsto q - k$ and $(x, y, z) \mapsto (x, -y, -z)$. Including both endpoints instead gives $q + 1$ points. The reversed planar coordinate order in [2, Section 5.2] gives an isometric unordered set: reindex $k \equiv pj \pmod{q}$ and use $p^2 \equiv (-1)^n \pmod{q}$; odd n also requires reflection of the second spherical coordinate.

The construction, pole upper bound and arithmetic product estimate are prior work. The contribution here is the all-pair spherical lower bound and its proof of the precisely stated conjecture. An extensive bounded primary-source audit found no earlier exact resolution in the sources inspected. It does not establish worldwide first priority or certify the conjecture’s openness as of 2026. The fate of the Brauchart–Dick in-preparation item cited in [1], and some final graphics editions, remains unverified; the accompanying priority note records the actual reading and access limits.

The accompanying verification package supplies exact finite arithmetic and certified rational chord controls, conventions and counterexample controls, source provenance, and reproduction instructions. These are checks supplementary to the proof, not a computational proof of an infinite theorem. AI tools were used extensively for derivation, literature comparison, computation, writing and independent adversarial verification. This is an unrefereed preprint; no human peer review or formal proof-assistant certification is claimed.

References

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